Calendar

Spring 2027

Last modified: 2026-10-23

The topics and dates of future lectures may change.

Class meets on Mondays, Tuesdays, Wednesdays and Fridays, 9:00–9:50am, in CF 420. The quarter's dates, the homework due dates and the exam dates are still to be set (TBD).

Monday to Wednesday are lectures from these notes. Friday is a lab session, for working on the homework in class.

Week	Topic
1	Course overview; machine learning review; tensors and automatic differentiation
2	Feedforward networks: layers, activations, output units and losses, representational power, backpropagation
3	Optimization: minibatch stochastic gradient descent, momentum, Adam, learning-rate schedules, initialization
4	Regularization and normalization: weight decay, dropout, early stopping, data augmentation, batch and layer normalization, residual connections
5	Convolutional networks: convolution arithmetic, pooling, classic architectures, transfer learning
6	Sequence models: embeddings, recurrent networks, LSTMs and GRUs, language modeling
7	Attention and Transformers: encoder-decoder attention, self-attention, positional encoding, pretraining
8	Generative models: autoencoders and variational autoencoders, generative adversarial networks, diffusion
9	Running deep learning experiments: hyperparameter types, sweeps, compute budgets, reproducibility
10	Applications and current topics: large language models, fine-tuning, speech and vision, ethics
Finals week	Optional final exam